Show that Okamoto's Signature scheme is EUF-CMA.

Let G be a cyclic group of prime order q with generators g and h
The secret key is $(\alpha, \beta) \in \mathbb{Z}_q^2$ and the public key is $X = g^{\alpha} h^{\beta}$

The okamoto identification protocol proves knowledge of (α, β) satisfying $X = g^{\alpha} h^{\beta}$.

& the Okamoto Signature Scheme is obtained via the Fiat-Shamir transform, replacing the verifier challenge by a hash function modeled as a random oracle: $\mathrm{Sign}(m) = (R, s_\alpha, s_\beta)$,

where $R = g^{k_\alpha} h^{k_\beta}$, $e = H(R, m)$, $s_\alpha = k_\alpha + e\alpha \pmod q$
$s_\beta = k_\beta + e\beta \pmod q$

Verification checks: $g^{s_\alpha} h^{s_\beta} \stackrel{?}{=} R \cdot X^e$

Our goal is to prove existential unforgeability under choosen message attack (EUF-CMA) against algebraic adversaries in the AGM-ROM, assuming DL hardness in G.

PROOF:

Suppose $\exists$ a PPT algebraic forger F who, given access to the signing oracle and the hash oracle, forges a valid signature $(m', R', s_\alpha', s_\beta')$ on some new message m'

We will construct an algorithm A that uses F to solve the discrete logarithm problem by extracting the secret exponents (α, β) from the forgery.

Simulation of oracle:

$\mathcal{A}$ receives the public key $X = g^{\alpha} h^{\beta}$ and must submit oracles for F.

Hash oracle H:

Initially empty. Whenever F queries $H(R, m)$:

- If (R, m) was queried before, return the same hash value.
- Otherwise, sample a uniform $e \leftarrow \mathbb{Z}_q$, store (R, m, e), return e.

Signing oracle for message m

$\mathcal{A}$ simulates the signature without knowing (α, β) by:

1) Choose random $s_{\alpha,i}\ s_{\beta,i} \in \mathbb{Z}_q$.
2) Compute $R_i := g^{s_{\alpha,i}} h^{s_{\beta,i}} X^{-e}$ where $e_i = H(R_i, m_i)$
3) If $H(R_i, m_i)$ was not programmed, program it now to equal e_i
4) Return the signature $(R_i, s_{\alpha,i}, s_{\beta,i})$

This simulation is consistent because

$$g^{s_{\alpha,i}} h^{s_{\beta,i}} = R_i X^{e_i}$$

Because F is algebraic (per AGM), every group element it outputs must be accompanied by an explicit algebraic expressions as a linear combination of known group elements g, h, X and R_i.

In particular, the forgery's commitment R' is

$$R' = g^{a_0} h^{a_1} \cdot X^{a_2} \cdot \prod_{i=1}^{q} R_i^{a_{i+2}}$$

for some exponents $a_0, a_1, a_2, \dots a_{q+2} \in \mathbb{Z}_q$

The forgery $(m', R', s_\alpha', s_\beta')$ satisfies the verification equation,

$$g^{s_\alpha'} \cdot h^{s_\beta'} \stackrel{?}{=} R' X^{e'} \quad \text{where } e' = H(R', m')$$

$$\Rightarrow g^{s_\alpha'} h^{s_\beta'} = g^{a_0} h^{a_1} X^{a_2} \prod_{i=1}^{q} R_i^{a_{i+2}} X^{e'}$$

$$= g^{a_0} h^{a_1} (g^\alpha h^\beta)^{a_2} \prod_{i=1}^{q} (g^{s_{\alpha,i}} h^{s_{\beta,i}} X^{-e_i})^{a_{i+2}} (g^\alpha h^\beta)^{e'}$$

$$= g^{a_0} h^{a_1} g^{\alpha a_2} h^{\beta a_2} \prod_{i=1}^{q} (g^{s_{\alpha,i}} h^{s_{\beta,i}} g^{-\alpha e_i} h^{-\beta e_i})^{a_{i+2}} (g^\alpha h^\beta)^{e'}$$

$$= g^{a_0 + \alpha a_2} \cdot h^{a_1 \cdot \beta a_2} \prod_{i=1}^{q} g^{(s_{\alpha,i} - \alpha e_i)(a_{i+2}) + \alpha e'} h^{(s_{\beta,i} - \beta e_i)(a_{i+2}) + \beta e'}$$

$$= g^{a_0 + \alpha a_2} h^{a_1 \beta a_2} g^{\sum_{i=1}^{q} (s_{\alpha,i} - \alpha e_i)(a_{i+2}) + \alpha e'} h^{\sum_{i=1}^{q} (s_{\beta,i} - \beta e_i)(a_{i+2}) + \beta e'}$$

Main Overview

- Proof of Existential Unforgeability under Chosen Message Attack (EUF-CMA) for Okamoto's Signature Scheme
- Proof conducted in Algebraic Group Model (AGM) and Random Oracle Model (ROM)
- Assumes Discrete Logarithm (DL) hardness in the group

Cryptographic Setup

- Group Characteristics:
- Cyclic group G of prime order q
- Generators: g and h
- Key Components:
- Secret key: (x, β)
- Public key: g ^ x h ^ β

Signature Scheme Details

- Based on Fiat-Shamir transform
- Signing Process:
- R = g ^ kx * h ^ kβ
- e = H(R, m)
- sx = kx * x * e (mod q)
- sp = kp * e * β (mod g)

Proof Strategy

- Goal: Prove EUF-CMA security against algebraic adversaries
- Approach:
- Use a PPT (Probabilistic Polynomial Time) forger F
- Construct an algorithm A to solve the discrete logarithm problem

Proof Mechanism

- Simulation of Oracles:
- Hash Oracle H
- Signing Oracle
- Key Constraints:
- Forger F must be algebraic
- Every group element must have an explicit algebraic expression

Proof Technique

- Extract secret exponents (x, β) from the forgery
- Verify that the forgery satisfies verification equation
- Demonstrate consistency of simulation

Verification Equation

- Must satisfy: g ^ sx' * h ^ sp' = R' * X ^ e'
- Where e' = H(R', m')

Cryptographic Assumptions

- Discrete Logarithm problem is hard
- Random Oracle Model (ROM)
- Algebraic Group Model (AGM)

Key Takeaway

The proof demonstrates that under the given cryptographic assumptions, the Okamoto Signature Scheme provides existential unforgeability against chosen message attacks.